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QUESTION BANK

Semester – III Examinatation – 2026

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MAJ-4

DATA SCIENCE

PYTHON IN DATA SCIENCE

UNIT I – Fundamentals of Data Science

2 Marks

1. Define Data Science.
2. What is Big Data?
3. What are the Five V's of Big Data?
4. What is Datafication?
5. What is a Data Scientist?
6. What is Statistical Inference?
7. What is Population and Sample?
8. Define Cross-sectional Data with an example.
9. What is Data Visualization?
10. What is Exploratory Data Analysis (EDA)?
11. What is Data Cleaning?
12. What is Modeling?
13. What is Data Science Process?
14. What is the role of a Data Scientist?

5 Marks

1. Explain the difference between Supervised and Unsupervised Learning with examples.
2. What is Feature Engineering in Machine Learning? Discuss some common techniques.
3. Explain the concept of Dimensionality Reduction. Discuss two common techniques.
4. What is Overfitting in a Machine Learning Model? How can it be prevented?
5. Discuss the concept of Bias-Variance Trade-off in model evaluation.
6. Write a short note on Exploratory Data Analysis.
7. Explain the steps involved in a typical Data Preprocessing Pipeline.
8. Explain the importance of Data Visualization in decision-making.
9. Differentiate between Supervised, Unsupervised and Reinforcement Learning.

UNIT II – Python Basics

2 Marks

1. What is Python?
2. Mention any two features of Python.
3. What is an Identifier?
4. What are Keywords?
5. What are Variables?
6. What are Data Types?
7. What are Operators?
8. What is Operator Precedence?
9. What is Indentation?
10. What are Comments?

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11. What is Type Conversion?
12. What is a Lambda Function?
13. What is the difference between == and is operators?
14. What is a Tuple in Python?
15. Explain Dictionary in Python with an example.
16. What is the difference between remove() and pop() in Lists?
17. What is the output of {1,2,3} & {3,4,5}?
18. What is a Python Interpreter?
19. What is the role of Python in Data Science?
20. What is an f-string?

5 Marks

1. Explain the difference between Lists and Tuples with examples.
2. Explain the properties of Python Sets.
3. Explain the difference between Supervised, Unsupervised and Reinforcement Learning. (Also relevant to Unit I.)
4. Explain the difference between Identity and Equality Operators.
5. Explain different Data Types in Python with examples.
6. Explain different types of Operators in Python with examples.
7. Explain Type Conversion with suitable examples.
8. Explain Dictionary and its operations with examples.
9. Explain the features and applications of Python in Data Science.
10. Explain Python Libraries and how to import them with examples.

UNIT III – Python Control Flow

2 Marks

1. What is Control Flow?
2. What is an if statement?
3. What is an if-else statement?
4. What is an elif statement?
5. What is a while loop?
6. What is a for loop?
7. What is a Nested Loop?
8. What is the use of break statement?
9. What is the use of continue statement?
10. What is the use of range() function?
11. What is the use of exit() function?
12. Differentiate between break and continue.

5 Marks

1. Explain the concept of Nested Loops with an example.
2. Explain different Control Flow Statements in Python with examples.
3. Explain if, if-else and elif statements with examples.
4. Differentiate between for loop and while loop with examples.
5. Explain break and continue statements with examples.
6. Explain the range() function with examples.
7. Find and explain the stepwise solution of the given Python expression: a = 2; b = 5; c = 10; a & b << 2 // 5 ** 2 + c ^ b.
8. Write a Python program to calculate the sum of digits of a number.

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UNIT IV – Python Functions & Strings

2 Marks

1. What is a Function in Python?
2. What are the different types of Functions?
3. What is Function Definition?
4. What is Function Calling?
5. What are Parameters and Arguments?
6. What is a Return Statement?
7. What are Default Parameters?
8. What are Keyword Arguments?
9. What are Command Line Arguments?
10. What is a String in Python?
11. What is String Concatenation?
12. What is String Slicing?
13. What is String Traversing?
14. What are Format Specifiers?
15. What are Escape Sequences?
16. What is a Raw String?
17. What is a Unicode String?
18. What is an f-string?

5 Marks

1. Describe the role of Return Statement in Functions.
2. Write a Python program to count the number of vowels and consonants in a string.
3. Write a Python program to calculate the sum of digits of a number using a function.
4. Explain different types of Functions in Python with examples.
5. Explain Function Definition, Function Calling and Passing Parameters with examples.
6. Explain Default Parameters and Keyword Arguments with examples.
7. Explain String Slicing and Concatenation with examples.
8. Explain different String Operations with examples.
9. Explain Escape Sequences with examples.
10. Explain String Formatting using f-strings with examples.
11. Write a Python program to reverse a string.
12. Write a Python program to check whether a string is a palindrome.

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